

AES-128 Simulation Report

Experiment 1 | Beyond Boolean | preserved behavioral simulation

1. Test configuration

Item	Recorded configuration
Design under test	aes_core; testbench instance uut
Testbench	tb_aes_core.v
Simulator record	Vivado XSim output saved in the native archive
Clock / timescale	10 ns period (100 MHz); 1 ns / 1 ps
Reset / start	Reset asserted initially, released after 100 ns; inputs and start applied after another 20 ns; start held for 10 ns.
Result collection	Wait for posedge done, delay 2 ns, compare ciphertext with the expected value.
Preserved coverage	One known-answer vector; no additional regression log located.

2. Observed archived result

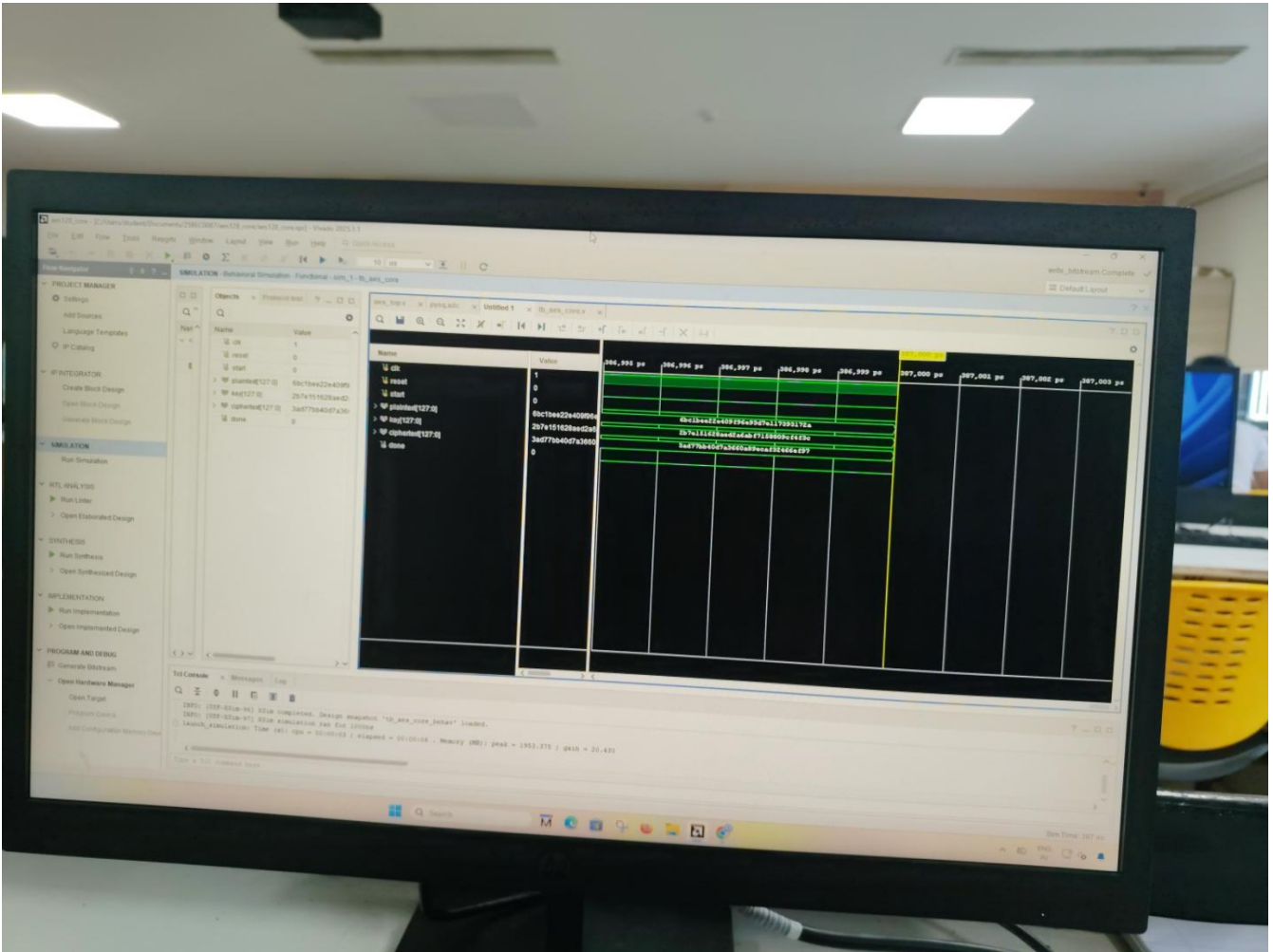
Signal	Archived hexadecimal value
Key	2b7e151628aed2a6abf7158809cf4f3c
Plaintext	6bc1bee22e409f96e93d7e117393172a
Expected ciphertext	3ad77bb40d7a3660a89ecaf32466ef97
Observed ciphertext	3ad77bb40d7a3660a89ecaf32466ef97

The original console prints >> **TEST PASSED** << and finishes at 387 ns. The result shown is read from the archived transcript; no rerun was performed for this document.

Handshake behavior

In aes_core.v, S_FINISHED assigns ciphertext and asserts done, then returns to S_IDLE. S_IDLE clears done on the next clock. This makes done a one-cycle pulse; a later GUI snapshot can show done=0 while the ciphertext remains correct.

3. Preserved waveform evidence



Unchanged original supplied photograph, also represented at smaller resolution in the uploaded project conversation. This historical snapshot is tightly zoomed near the end of a run.

The photograph shows the plaintext, key and ciphertext buses, but it is a limited presentation view. Interpret the photographed reset/done state as that instant only; it is not a full transaction timing plot. The archived transcript is the direct evidence of the saved comparison result.

Full-transaction capture

The Simulation/ folder includes the original tb_aes_core_behav.wdb. Open the database in Vivado static simulation, add the relevant signals, select hexadecimal bus radix and zoom to the full transaction. Replace waveform.png with a clearer capture if available. If the database lacks sufficient recorded signals, rerun the preserved testbench and save the new log with the corresponding image.

Verification boundary

Behavioral functional simulation does not establish routed timing closure. The implementation constraint is 125 MHz, while this testbench uses 100 MHz. The preserved routed report has WNS -0.104 ns and four setup-failing endpoints.

4. Original transcript

The text below is taken from the preserved simulate.log. Long lines are wrapped only for this PDF; Simulation/transcript.txt retains the original bytes.

Time resolution is 1 ps

```
=====
AES-128 ENCRYPTION TEST

KEY : 2b7e151628aed2a6abf7158809cf4f3c
PLAINTEXT : 6bc1bee22e409f96e93d7e117393172a
CIPHERTEXT : 3ad77bb40d7a3660a89ecaf32466ef97
EXPECTED : 3ad77bb40d7a3660a89ecaf32466ef97

STATUS : >> TEST PASSED <<
=====
```

```
$finish called at time : 387 ns : File "C:/Users/student/Documents/25BEC0087/aes128_
core/aes128_core.srscs/sim_1/new/tb_aes_core.v" Line 67
```

5. Conclusions and limits

The saved single-vector test passed and produced the expected 128-bit ciphertext. The recovered evidence does not contain broader vector coverage, randomized tests, decryption tests, a new simulator run or a timing-closed 125 MHz implementation. Those claims should not be added without corresponding results.

Included evidence files

Path	Provenance
Testbench/tb_aes_core.v	Unchanged original testbench from the archive.
RTL/aes_core.v	Unchanged original AES RTL from the archive.
Simulation/transcript.txt	Original XSim simulate.log, copied byte-for-byte.
Simulation/tb_aes_core_behav.wdb	Original native waveform database.
Simulation/waveform.png	Original uploaded waveform photograph.